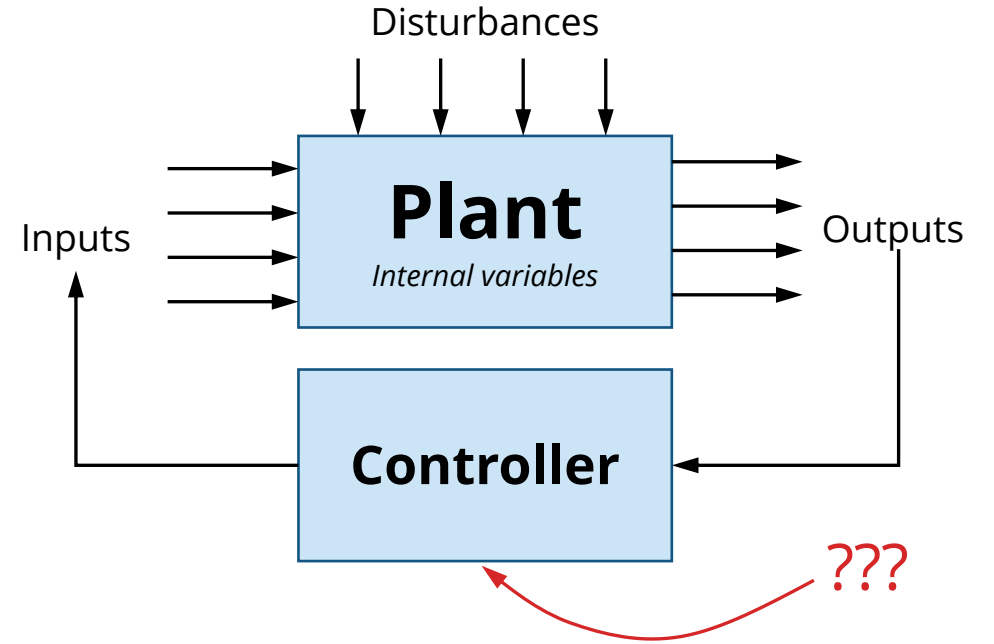




AA 513 Multivariable Control

Lecture 06 // Week 6

Instructor: Prof. Karen Leung

Announcements

- Exercises 9, 10, 11, 12 out (soon)
- Self-assessments due this week
- Check-in sessions by appointment to discuss research papers and tutorial topics.
- Research paper presentation
 - Follow PACES framework
 - Shared slide deck released
- Tutorial submission
 - **Submission:** Upload video to shared folder
 - **Peer review:** You will be assigned videos to review. You are welcome to review more if you like.

Week 10 schedule

Schedule

- 6 minutes per talk + 2 minute questions

Session 1			Session 2			Session 3		
#	Time	Name	#	Time	Name	#	Time	Name
1	6:00PM		8	7:00PM		15	8:00PM	
2	6:08PM		9	7:08PM		16	8:08PM	
3	6:16PM		10	7:16PM		17	8:16PM	
4	6:24PM		11	7:24PM		18	8:24PM	
5	6:30PM		12	7:30PM		19	8:30PM	
6	6:38PM		13	7:38PM		20	8:38PM	
7	6:46PM		14	7:46PM		21	8:46PM	
						22	8:54PM	

Last time

- Guest lecture by Dr Liam Krause, topic on Markov Decision Process and algorithms for finding optimal policies
 - **MDP:** More general framework for modeling systems that evolve over time
 - Assumes transition *probabilities*
 - Expectation over transition probabilities
 - **Value iteration:** DP until convergence
 - **Policy iteration:** Iterate between policy evaluation & policy improvement
 - **POMDPs:** *Partially observable* MDPs. Keep a *belief* over your state.
 - Expectation over transition AND belief probabilities
 - **Street-fighting techniques:** Value function approximations, MCTS (local search)

State Space $\mathcal{S}$

Action Space $\mathcal{A}$

Transition Model $T(s' | s, a)$

Reward Function $R(s, a)$

This week

- Continuous time: Hamilton-Jacobi-Bellman equation
- Linear quadratic regulator
- No lecture next week (Veteran's Day)
 - Will release recording

Exercises

- Ex9 Stochastic Bellman equation
- Ex10: Finite horizon LQR
- Ex11: Infinite horizon LQR
- Ex12: Tracking LQR

Continuous-time, continuous state & control

- Consider cost $J_N(\mathbf{x}(T)) + \int_0^T g(\mathbf{x}(t), \mathbf{u}(t), t) dt$
- Cast as discrete time problem with timestep ϵ
 - $J_{\text{discrete}}(\mathbf{x}_t, \mathbf{u}_t, t) = \int_t^{t+\epsilon} g(\mathbf{x}(\tau), \mathbf{u}(\tau), \tau) d\tau \approx \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t)$, for small ϵ

$$\underline{V^*(\mathbf{x}(t), t)} = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + V^*(\mathbf{x}(t + \epsilon), t + \epsilon)$$

$$0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + \underline{V^*(\mathbf{x}(t + \epsilon), t + \epsilon)} - V^*(\mathbf{x}(t), t)$$

$$0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \epsilon g(\mathbf{x}(t), \mathbf{u}(t), t) + \underbrace{V^*(\mathbf{x}(t), t) + \nabla_x V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \epsilon + \frac{\partial V}{\partial t} \epsilon + O(\epsilon^2)}_{\text{Taylor expansion}} - V^*(\mathbf{x}(t), t)$$

$$\div \epsilon \quad 0 = \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \underbrace{g(\mathbf{x}(t), \mathbf{u}(t), t)}_{\text{no dependence on } u} + \frac{1}{\epsilon} (\nabla_x V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \epsilon + \frac{\partial V}{\partial t} \epsilon + O(\epsilon^2))$$

$$0 = \lim_{\epsilon \rightarrow 0} \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla_x V^*(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) + \frac{\partial V}{\partial t} + O(\epsilon)$$

$$0 = \underbrace{\frac{\partial V}{\partial t}(\mathbf{x}(t), t)}_{\text{no dependence on } u} + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \left(g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t) \right)$$

(Bellman eq. - time)
move LHS to RHS

Hamilton-Jacobi-Bellman equation

Consider cost $g_N(\mathbf{x}(T)) + \int_0^T g(\mathbf{x}(t), \mathbf{u}(t), t) dt$

PDE!

$$\frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in U(\mathbf{x}(t))} \{g(\mathbf{x}(t), \mathbf{u}(t), t) + \nabla V(\mathbf{x}(t), t)^T f(\mathbf{x}(t), \mathbf{u}(t), t)\} = 0$$

Terminal condition: $V(\mathbf{x}, T) = g_N(\mathbf{x})$ terminal condition

Solve PDE backward in time. But generally not easy to solve ☹

What have we just done?

- Derived the key equations describing the optimal solution to a sequential decision-making problem
 - Under some assumptions, these are necessary and sufficient conditions for optimality
- Solving exactly is hard / intractable
- Instead, perform “**approximate** dynamic programming” and **function approximators** for value function

Can we ever hope to solve these equations exactly?!

- Yes! Under strict assumptions on the functional form of dynamics, cost, and value function.
- This is known as the linear quadratic regulator!

The discrete-time finite horizon LQR problem

Linear $\nearrow$ quadratic $\nwarrow$ regulator $\nwarrow$

Regulate to zero, want state and control to be small

$$\min_{\substack{\mathbf{u}_{0:T-1} \\ \text{sequence of control}}} \underbrace{\mathbf{x}_T^T Q_T \mathbf{x}_T}_{\text{terminated cost}} + \sum_{t=0}^{T-1} \underbrace{\mathbf{x}_t^T Q_t \mathbf{x}_t}_{\text{running cost}} + \underbrace{\mathbf{u}_t^T R_t \mathbf{u}_t}_{\text{regulator}} \quad \text{Quadratic cost} \quad (1)$$

$$\text{subject to } \mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t, \quad t = 0, \dots, T-1 \quad (2)$$

Linear dynamics

$$\mathbf{x}_0 = \mathbf{x}_{\text{current}} \quad (3)$$

Initial state

* Not really a constraint since state is not an optimization variable. More to just “anchor” the problem to a starting state.

Some notes about the LQR problem

$$\begin{aligned} \min_{\mathbf{u}_{0:T-1}} \quad & \mathbf{x}_T^T \underline{Q_T} \mathbf{x}_T + \sum_{t=0}^{T-1} \mathbf{x}_t^T \underline{Q_t} \mathbf{x}_t + \mathbf{u}_t^T \underline{R_t} \mathbf{u}_t \\ \text{subject to} \quad & \mathbf{x}_{t+1} = \underline{A_t} \mathbf{x}_t + \underline{B_t} \mathbf{u}_t, \quad t = 0, \dots, T-1 \\ & \mathbf{x}_0 = \mathbf{x}_{\text{current}} \end{aligned}$$

- No constraints! (aside from dynamics)
- Cost and dynamics can be **time-varying**
- Is it always possible to regulate to zero? (i.e., stabilize system?)

- Can we even control it? *at the very least, want system to be stabilizable. (uncontrollable modes to be stable).*
- What if there are uncontrollable/unobservable modes? *detectable as well.*

- Conditions on the cost matrices.

$$\begin{aligned} Q_t &\geq 0 \text{ (PSD)} & Q_t &= Q_t^T, \text{ erg}(Q_t) \geq 0 \\ Q_T &\geq 0 \text{ (PSD)} & Q_T &= Q_T^T \end{aligned}$$

$$\begin{aligned} R_t &> 0 \text{ (PD)} & R_t &= R_t^T \\ \text{erg}(R_t) &> 0 \end{aligned}$$

How to solve the LQR problem?

$$\begin{aligned} \min_{\mathbf{u}_{0:T-1}} \quad & \mathbf{x}_T^T Q_T \mathbf{x}_T + \sum_{t=0}^{T-1} \mathbf{x}_t^T Q_t \mathbf{x}_t + \mathbf{u}_t^T R_t \mathbf{u}_t \\ \text{subject to} \quad & \mathbf{x}_{t+1} = A_t \mathbf{x}_t + B_t \mathbf{u}_t, \quad t = 0, \dots, T-1 \\ & \mathbf{x}_0 = \mathbf{x}_{\text{current}} \end{aligned}$$

- Is the problem convex? *Yes - can use CVXPY!*

- Can we solve it in closed-form?

can we have $\mathbf{u}_t^ = \mathbf{K}(x_t)$*

Recall: Bellman equation

$$V^*(\mathbf{x}_t, t) = \min_{\mathbf{u} \in \mathcal{U}(\mathbf{x}_t)} g(\mathbf{x}_t, \mathbf{u}_t, t) + V^*(f(\mathbf{x}_t, \mathbf{u}_t, t), t+1), \quad V^*(\mathbf{x}_T, T) = \underline{g_T(\mathbf{x}_T)}$$

$$g_T(\mathbf{x}_T) = \mathbf{x}_T^T \mathbf{Q}_T \mathbf{x}_T$$

$$f(\mathbf{x}_t, \mathbf{u}_t, t) \Rightarrow \mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t$$

$$g(\mathbf{x}_t, \mathbf{u}_t, t) \Rightarrow \mathbf{x}_t^T \mathbf{Q} \mathbf{x}_t + \mathbf{u}_t^T \mathbf{R} \mathbf{u}_t$$

$$\mathbf{u} \in \mathcal{U}(\mathbf{x}_t) \quad \text{no constraints!}$$

$$V^*(\mathbf{x}_t, t) = \min_{\mathbf{u}_t} \mathbf{x}_t^T \mathbf{Q} \mathbf{x}_t + \mathbf{u}_t^T \mathbf{R} \mathbf{u}_t + V^*(\mathbf{A}\mathbf{x}_t + \mathbf{B}\mathbf{u}_t, t+1)$$

$$V^*(\mathbf{x}_T, T) = \mathbf{x}_T^T \mathbf{Q}_T \mathbf{x}_T.$$

time invariant

Simplifying assumptions: Not time-varying

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terminal condition $V(x_T, T) = x_T^T Q_T x_T$ (it is quadratic)

assume $V(x_t, t) = x_t^T P_t x_t$, then $P_T = Q_T$ $x_{t+1} = A x_t + B u_t$

$$V(x_{T-1}, T-1) = \min_{u_{T-1}} x_{T-1}^T \underset{T-1}{Q} x_{T-1} + u_{T-1}^T \underset{T-1}{R} u_{T-1} + (\underset{T-1}{A} x_{T-1} + \underset{T-1}{B} u_{T-1})^T P_T (\underset{T-1}{A} x_{T-1} + \underset{T-1}{B} u_{T-1}) \quad \leftarrow$$

$$= \min_{u_{T-1}} x_{T-1}^T \underset{T-1}{Q} x_{T-1} + u_{T-1}^T \underset{T-1}{R} u_{T-1} + x_{T-1}^T \underset{T-1}{A}^T P_T \underset{T-1}{A} x_{T-1} + \underbrace{2 u_{T-1}^T B^T P_T A x_{T-1}} + u_{T-1}^T B^T P_T B u_{T-1}$$

$$= \min_{u_{T-1}} \underbrace{x_{T-1}^T (Q + A^T P_T A) x_{T-1}}_{\text{no dependence on } u} + \underbrace{u_{T-1}^T (R + B^T P_T B) u_{T-1}}_{\text{quadratic in } u} + \underbrace{2 u_{T-1}^T B^T P_T A x_{T-1}}_{\text{linear in } u} \quad (*)$$

$$\left[\frac{d}{dx} (x^T a) = a \text{ (column)}, \quad \frac{d}{dx} (x^T A x) = 2 A x \text{ (column)} \right]$$

$$\frac{d}{du_{T-1}} (*) = 2(R + B^T P_T B) u_{T-1} + 2 B^T P_T A x_{T-1} = 0$$
$$\Rightarrow \boxed{u_{T-1}^* = - (R + B^T P_T B)^{-1} B^T P_T A x_{T-1}} = K_{T-1} x_{T-1}$$

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So we have $u_{T-1}^* = -(R+B^T P_T B)^{-1} B^T P_T A x_{T-1} \Rightarrow \text{Linear controller!}$

$$\text{set } K_{T-1} = -(R+B^T P_T B)^{-1} B^T P_T A$$

$$V(x_{T-1}, T-1) = x_{T-1}^T (Q + A^T P_T A) x_{T-1} + \underbrace{x_{T-1}^T K_{T-1}^T (R+B^T P_T B) K_{T-1} x_{T-1}} + 2 x_{T-1}^T K_{T-1}^T B^T P_T A x_{T-1}$$

$$\begin{aligned} K_{T-1}^T (R+B^T P_T B) K_{T-1} &= \overbrace{A^T P_T B (R+B^T P_T B)^{-1}} (\cancel{\sim}) (\cancel{\sim}) \underbrace{B^T P_T A}_{T(A^T)^T = (A^T)^{-1}} \\ &= \overbrace{A^T P_T B (\cancel{\sim})}^{-1} B^T P_T A \\ &= \underline{-K_{T-1}^T B^T P_T A} \end{aligned}$$

$$\begin{aligned} &\Rightarrow x_{T-1}^T (Q + A^T P_T A) x_{T-1} + \cancel{x_{T-1}^T K_{T-1}^T B^T P_T A x_{T-1}} + 2 x_{T-1}^T K_{T-1}^T B^T P_T A x_{T-1} \\ &= x_{T-1}^T (Q + P_T A) x_{T-1} + \cancel{x_{T-1}^T K_{T-1}^T B^T P_T A x_{T-1}} \end{aligned}$$

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$$\uparrow = x_{T-1}^T \left(Q + A^T P_T A + K_{T-1}^T B^T P_T A \right) x_{T-1} \quad \text{Also quadratic!}$$

$V(x_{T-1}, T-1)$

$$V(x_{T-1}, T-1) = x_{T-1}^T P_{T-1} x_{T-1}$$

$$\text{where } P_{T-1} = Q + A^T P_T A + K_{T-1}^T B^T P_T A$$

so we can generalise this. for all t .

$$\boxed{\begin{aligned} P_{t-1} &= Q + A^T P_t A - A^T P_t B (R + B^T P_t B)^{-1} B^T P_t A \\ K_{t-1} &= - (R + B^T P_t B)^{-1} B^T P_t A \end{aligned}}$$

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Discrete-time finite horizon LQR (time-invariant)

1. set $P_T = Q_T$

2. for $t = N, \dots, 1$,

$$\underline{P_{t-1}} = Q + A^T P_t A - A^T P_t B (R + B^T P_t B)^{-1} B^T P_t A$$

$$\underline{K_{t-1}} = -(R + B^T P_t B)^{-1} B^T P_t A$$

3. for $t = 0, \dots, N - 1$, $\underline{u_t^*} = K_t \mathbf{x}_t$

Discrete-time finite horizon LQR (time-varying)

What changes?

1. set $P_T = Q_T$

2. for $t = N, \dots, 1$,

$V(x_{t-1}, t-1)$

$$\underline{P_{t-1}} = \underline{Q_{t-1}} + \underline{A_{t-1}^T} \underline{P_t} \underline{A_{t-1}} - \underline{A_{t-1}^T} \underline{P_t} \underline{B_{t-1}} (\underline{R_{t-1}} + \underline{B_{t-1}^T} \underline{P_t} \underline{B_{t-1}})^{-1} \underline{B_{t-1}^T} \underline{P_t} \underline{A_{t-1}}$$
$$K_{t-1} = -(\underline{R_{t-1}} + \underline{B_{t-1}^T} \underline{P_t} \underline{B_{t-1}})^{-1} \underline{B_{t-1}^T} \underline{P_t} \underline{A_{t-1}}$$

3. for $t = 0, \dots, N - 1$, $\mathbf{u}_t^* = K_t \mathbf{x}_t$

Discrete-time **infinite** horizon LQR (time-invariant)

What changes? as $T \rightarrow \infty$ $P_t = P_{t-1} = P_\infty = P$

1. set $P_T = Q_T$

2. for $t = N, \dots, 1$,

Discrete Algebraic Riccati Equation (DARE)

$$P_{\text{min}} = Q + A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A$$

$$K_{t-1} = -(R + B^T P B)^{-1} B^T P A$$

3. for $t = 0, \dots, N - 1$, $\mathbf{u}_t^* = K_t \mathbf{x}_t$

What about the continuous-time setting?

- Same process but with the HJB equation!

$$0 = \frac{\partial V}{\partial t}(\mathbf{x}(t), t) + \min_{\mathbf{u}(t) \in \mathcal{U}(\mathbf{x}(t))} \underline{g(\mathbf{x}(t), \mathbf{u}(t), t)} + \underline{\nabla V(\mathbf{x}(t), t)^T} \underline{f(\mathbf{x}(t), \mathbf{u}(t), t)}$$

$$\min_{\mathbf{u}(\cdot)} \overset{T}{\mathbf{x}(T)} Q_T \mathbf{x}(T) + \int_0^T \overset{T}{\mathbf{x}(t)} Q(t) \mathbf{x}(t) + \overset{T}{\mathbf{u}(t)} R(t) \mathbf{u}(t) \quad (1)$$

$$\text{subject to } \dot{\mathbf{x}} = A(t)\mathbf{x}(t) + B(t)\mathbf{u}(t), \quad t = [0, T] \quad (2)$$

$$\mathbf{x}(0) = \mathbf{x}_{\text{current}} \quad (3)$$

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$$0 = \frac{\partial V}{\partial t} + \min_u x^T Q x + u^T R u + \nabla V(x,t)^T \underline{\underline{[Ax + Bu]}}$$

assume $V(x,t) = x^T(t) P(t) x(t) = \underline{\underline{x^T P(t) x}}$

$$\frac{\partial V}{\partial t} = x^T \dot{P}(t) x, \quad \nabla V(x,t) = \frac{\partial}{\partial x} V(x,t) = \frac{\partial}{\partial x} (x^T P(t) x) = 2 P x$$

$$0 = \underbrace{x^T \dot{P} x}_{\text{Scalar}} + \min_u \underbrace{x^T Q x}_{\text{Scalar}} + \underbrace{u^T R u}_{\text{Scalar}} + 2 x^T P A x + \boxed{\underline{\underline{(2 x^T P B u)^T}}}$$

$$0 = x^T \dot{P} x + \min_u x^T Q x + \underline{\underline{u^T R u}} + 2 x^T P A x + \underline{\underline{2 u^T B^T P x}}$$

$$\frac{\partial}{\partial x} (x^T a) = a$$

$$\frac{\partial}{\partial x} (x^T A x) = 2 A x$$

$$\frac{\partial}{\partial u} (\min) = 2 R u + 2 B^T P x = 0$$

$$R u = -B^T P x$$

$$\Rightarrow u^* = -R^{-1} B^T P x$$

$$u^* = \underbrace{-R^{-1}}_{[m \times m]} \underbrace{x^T P B}_{[1 \times n]}$$

$$\underline{\underline{-R^{-1} B^T P x}}$$

$[m \times m][n \times n][n \times n][n \times 1] \checkmark$

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$$u^*(t) = -R^{-1}B^T P(t)x(t)$$

$$0 = x^T \dot{P}x + x^T Qx + \underbrace{x^T P B R^{-1} R R^T B^T P x}_{\cancel{\text{cancel}}} + 2x^T P A x - \underbrace{2x^T P B R^{-1} B^T P x}_{\cancel{\text{cancel}}}$$

$$0 = x^T \dot{P}x + x^T Qx + 2x^T P A x - x^T P B R^{-1} B^T P x$$

$$0 = x^T \left(\dot{P} + Q + 2PA - PBR^{-1}B^T P \right) x$$

take the symmetric part of $\uparrow$

$\dot{P} = \dot{P}^T$ $Q = Q^T$ $PBR^{-1}B^T P$ is symmetric is $2PA$ symmetric? generally not.

recall $M = \underbrace{\frac{M+M^T}{2}}_{\text{sym.}} + \underbrace{\frac{M-M^T}{2}}_{\text{skew sym.}}$ $x^T M x^T \equiv x^T \left(\frac{M+M^T}{2} \right) x$

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$$\begin{aligned}\text{sym}(2PA) &= \frac{2PA + (2PA)^T}{2} \\ &= PA + A^T P\end{aligned}$$

$$0 = \underline{x}^T \left(\dot{P} + Q + \underbrace{PA + A^T P}_{\text{Riccati ODE}} - PBR^{-1}B^T P \right) \underline{x}$$

$$\therefore \text{we must have } \boxed{-\dot{P} = Q + PA + A^T P - PBR^{-1}B^T P} \quad \text{ODE!}$$

$P(T) = Q_T$

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Continuous-time finite horizon LQR (time-invariant)

1. set $P(T) = Q_T$

2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^* = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Continuous-time finite horizon LQR (time-varying)

What changes?

1. set $P(T) = Q_T$

2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^* = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Continuous-time **infinite** horizon LQR (time-invariant)

What changes?

1. set $P(T) = Q_T$

2. Solve Riccati differential equation *backward* in time,

$$-\dot{P}(t) = A^T P(t) + P(t)A - P(t)BR^{-1}B^T P(t) + Q, \quad P(T) = Q_T$$

3. Optimal control $\mathbf{u}(t)^* = K(t)\mathbf{x}(t)$, $K(t) = -R^{-1}B^T P(t)$

Other variations

- Regulating to $(\mathbf{x} - \mathbf{x}_g)^T Q (\mathbf{x} - \mathbf{x}_g) + (\mathbf{u} - \mathbf{u}_g)^T R (\mathbf{u} - \mathbf{u}_g)$
 - Cost has linear terms
 - Value function takes on the form: quadratic + linear + constant
 - Recursion updates for all terms
 - Controller is affine
- Tracking LQR $(\mathbf{x}_t - \tilde{\mathbf{x}}_t)^T Q (\mathbf{x}_t - \tilde{\mathbf{x}}_t) + (\mathbf{u}_t - \tilde{\mathbf{u}}_t)^T R (\mathbf{u}_t - \tilde{\mathbf{u}}_t)$
 - Have a desired trajectory to follow
- Iterative LQR
 - Iteratively approximate non-LQR problem as LQR problem and solve several times

Tracking LQR

- Consider desired state and control sequence
 - $(\tilde{\mathbf{x}}_0, \tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_{T-1}), (\tilde{\mathbf{u}}_0, \tilde{\mathbf{u}}_1, \dots, \tilde{\mathbf{u}}_{T-1})$
- Consider *error* states, controls, and dynamics
 - $\delta \mathbf{x}_t = \mathbf{x}_t - \tilde{\mathbf{x}}_t$
 - $\delta \mathbf{u}_t = \mathbf{u}_t - \tilde{\mathbf{u}}_t$
 - $\delta \dot{\mathbf{x}}_t = \dot{\mathbf{x}}_t - \dot{\tilde{\mathbf{x}}}_t = A_t \mathbf{x}_t + B_t \mathbf{u}_t - (A_t \tilde{\mathbf{x}}_t + B_t \tilde{\mathbf{u}}_t) = A_t \delta \mathbf{x}_t + B_t \delta \mathbf{u}_t$
- Want error $\rightarrow 0$
 - Apply standard LQR on *error* states, controls, and dynamics
 - $\delta \mathbf{u}_t^* = K_t \delta \mathbf{x}_t \Rightarrow \mathbf{u}_t^* = K_t \delta \mathbf{x}_t + \tilde{\mathbf{u}}_t$